Spring AI: Beginner to Guru

Generative AI with Spring

Types of Artificial Intelligence

Types of Artificial Intelligence

- **Artificial Narrow AI** - Also known as Weak AI
  - This is the only AI that exists today
  - Can be trained to perform a single or narrow task
- **General AI** - still theoretical - Artificial General Intelligence or AGI. Also known as Strong AI
  - Can use previous learnings to accomplish new tests in a different context
  - Can do any intellectual task that a human can
  - Estimates range from months to years for when AGI

Types of Artificial Intelligence

- **Super AI** - strictly theoretical
- If ever developed
  - Would think, reason, learn and make judgements
  - Cognitive Abilities would surpass humans
  - Would evolve beyond human sentiments and emotions
  - Would have beliefs and desires of its own

Types of Narrow AI

- **Reactive Machine AI** - AI Systems without memory
- Designed for specific task
- Cannot remember previous outcomes or decisions
- Examples
  - **IBM Deep Blue** - Chess playing super computer
  - **Netflix Recommendation Engine** - Viewing recommendations based on viewing history

Types of Narrow AI

- **Limited Memory AI** - Can recall past events and outcomes and can monitor situations over time
  - As it is trained on more data over time, it can improve performance
- Examples of Limited AI
  - **Generative AI** - Tools such as ChatGPT, Gemini, Gork
  - **Virtual Assistants and chatbots** - Siri, Alexa, Google Assistant
  - **Self Driving Cars** - Limited Memory AI used to pilot the vehicle

Generative AI

- Generative AI is a type of AI which focuses on generating content such as text, images, video, or audio
- Also can learn patterns and structures of large data sets
- Problems
  - Susceptible to biases present in training data
  - Requires large amounts of computational power
  - Lacks transparency making it difficult to understand its reasoning

Deterministic AI

- Type of AI that makes decisions based on pre-defined rules and procedures
  - Ensures consistent and accurate decision making
  - Provides transparency and traceability
- Problems:
  - May be inflexible and unable to adapt to new situations
  - Requires manual updates to accommodate changes

Reactive AI

- Type of AI which focuses on reacting to specific situations
  - Also known as Rule Based AI
- Does not have an internal model. Decisions are made solely on the input of the situation
- Problems
  - Lacks ability to learn from past experiences or adapt to new situations
  - May require extensive rule development

Comparison

Feature	Generative AI	Deterministic AI	Reactive AI
Consistency	Moderate	High	High
Transparancy	Low	High	Moderate
Adaptability	High	Low	Low
Learning Ability	High	None	None
Compliance Assurance	Moderate	High	Moderate
Complexity Handling	High	Moderate	Low
Resource Requirements	High	Low	Low
Real Time Decisioning	Moderate	Moderate	High
Domain Specific Expertise	Moderate	High	High

SPRING FRAMEWORK GURU